

Stem Cells and Regeneration

Sérgio F genetic report
2 of October 2026 (local time)

Name	Sérgio F
Focus	longevity and immortality (84 aging mechanisms)
Data	Crude DNA (chip) + local imputation (only DR2 ≥ 0.8)
Content	7 Curated SNPs · 40 loci of GWAS · 3 polygenic scores · 10 045 pathogenic variants of ClinVar in 32 genes

What is: Sérgio F predisposition to exhaustion and clonal expansion of blood stem cells: clonal hematopoiesis (CHIP), mosaic loss of Y, haplotype JAK2 46/1, TCL1A, SH2B3, TERT and telomeric reserve, more rare variants in 32 genes of spinal cord failure and myeloid predisposition.
Privacy: the analysis was done locally; the genotypes never left the computer. Public databases were only sent SNP identifiers.

Important: genetic predispositions, not a diagnosis. The conclusions come only from the genes (without analysis or exams). All practical ideas are suggestion (not applied), to decide with the doctor.

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1. Abstract: favourable and unfavourable

Favourable	Unfavourable
• Without the haplotype JAK2 46/1 (rs12343867 TT): less predisposition to acquire the JAK2 V617F mutation. • SH2B3 rs3184504 CC: without the Trp262 allele that releases the stem cell proliferation brake. • Y loss: Score at the 86th percentile of dosage (less predicted loss); CHIP: score at the 30th percentile, 7 loci favourable (e.g. RUNX1). • Non carrier CHEK2 I157T or TP53 rs78378222. • ClinVar: without plausible variants in 32 genes; SBDS and ATM in homozygosity are artifacts.	• Genetically shorter telomeres (score at the 25th percentile): less replicative reserve of stem cells. • TERT rs2736100 CC (commitment): extends telomeres, but increases the risk of myeloproliferative neoplasms and clonal expansions. • TCL1A rs2887399 GG: without the T allele that stops the expansion of clones DNMT3A/TET2 (typical; is the most frequent genotype). • 2 unfavourable CHIP loci (CD164, ITPR2).

Reading for immortality: Blood stem cells have a low to typical predisposition for clonal expansions (without JAK2 46/1, favourable SH2B3, favourable CHIP and loss of Y), with telomeric reserve slightly below average. For immortality, the objective is to preserve the reserve (less inflammation and damage) without stimulating proliferation (nothing of telomerase activators or growth factors without indication).

2. Method, chain and trust

- **Genotypes:** 15 read directly on the chip (GRCh38, strand +); 28 locally imputed (Beagle 5.5, full genome reference panel 30x, only DR2 ≥ 0.8); 4 without data ("SNP not found in the data").
- **Strand:** All alleles were aligned to the GRCh38 strand + and verified against alleles and frequencies of the reference panel. 0 alleles of the literature were in the strand coding – and were converted (complement); 5 palindromic SNPs (A/T or C/G) were resolved by the allelic frequency; the non-resolvable palindromic SNPs were out of the scores.
- **Classification:** compares the copies of the effect allele with the average of the reference population (2 × frequency). Difference ≥ 0.5 copies → Favourable or Unfavourable; lower → Typical. Not having a rare variant of risk, or a common variant of strong effect, counts as Favourable. An allele with opposite effects (e.g. longer telomeres but higher risk of cancer) appears as Commitment in both lists.
- **Trust:** chip > imputation (DR2 ≥ 0.8; estimated accuracy of ~95–99% in common variants, lower in rare ones). Evidence: Strong (replicated in large/functional GWAS), Moderate, Weak (small or inconsistent studies).
- **Polygenic scores:** weighted sum of alleles (published coefficients); percentile calculated against the reference population with exactly the same variants.
- **Sources of interpretation:** dbSNP/Ensembl, ClinVar, GWAS Catalog, PGS Catalog and articles cited in the text.

3. Polygenic scores

Score	Percentile	Variants used	Interpretation	Note
Clonal hematopoiesis (21 loci, Wen 2025) G_CHIP · calculated in this report	30	21 of 21 (100%)	Typical	Typical, at the lower limit (percentile 30.5).
Y-chromosome dosage (less mosaic loss; 14 loci) G_MLOY · calculated in this report	86	14 of 14 (100%)	Favourable	High percentile : less predicted Y loss (favourable).
Telomere length (87 loci) G_TL · calculated in this report	25	86 of 87 (99%)	Unfavourable	Low percentile = less replicative reserve (but also less risk of myeloid clones).

the 50th percentile : median of the reference population. For risk scores, high percentile = higher risk; for "pro" scores (e.g. longer telomeres, longevity), high percentile = more of the trait. Typical = percentile between 30 and 70. The scores explain only a small part of the variation and are not diagnostic.

4. Favourable

Gene/score	SNP	Genotype	Source · evidence	Effect
JAK2	rs12343867 haplotype 46/1 (GGCC)	TT 0× C	chip Strong	Without the C-risk allele. The C allele marks the haplotype 46/1, which increases 3–4 times the probability of acquiring the JAK2 V617F mutation and myeloproliferative neoplasia. Mechanisms: Clonal hematopoiesis (CHIP); Exhaustion of Stem Cells
TERT	rs2736100 Intron 2 (TERT)	CC 2× C	chip Strong	Allele C: longer telomeres and more storage of stem cells. Mechanisms: Exhaustion of Stem Cells; Clonal hematopoiesis (CHIP); Hayflick's Telomeres and Limit
SH2B3	rs3184504 Trp262Arg (LNK)	CC 0× T	chip Moderate	Without the T-risk allele. The T allele (Trp262) reduces the function of LNK, a brake on the proliferation of hematopoietic stem cells: more blood cells, more myeloproliferative neoplasms, coronary disease and autoimmunity. Mechanisms: Exhaustion of Stem Cells; Clonal hematopoiesis (CHIP); Inflammaging
CHEK2	rs17879961 I157T	AA 0× G	chip Moderate	Without the risk allele G. Rare variant associated with clonal hematopoiesis and chronic lymphocytic leukemia. Mechanisms: Clonal hematopoiesis (CHIP); DNA Damage and DNA Repair Deficit
TP53	rs78378222 polyadenylation signal	TT 0× G	chip Moderate	Without the risk allele G. Rare allele associated with more mosaic loss of Y chromosome and several cancers. Mechanisms: Somatic / Tissue mosaicism; Clonal hematopoiesis (CHIP)
polygenic	G_MLOY Y-chromosome dosage (less mosaic loss; 14 loci)	the 86th percentile	14 variants Score	High percentile : less predicted Y loss (favourable). Mechanisms: Somatic / Tissue mosaicism; Clonal hematopoiesis (CHIP)
TRIM59-IFT80, POGLUT3, UBE2G1, RUNX1, ZNF318, ACTG1P25	Clonal hematopoiesis (CHIP) 7 of 23 loci		GWAS GWAS	7 loci in which the genotype of Sérgio F deviates from the mean of the reference population in the favourable sense (favourable = less risk of CHIP). Mechanisms: Clonal hematopoiesis (CHIP); Exhaustion of Stem Cells; Somatic Mutation Accumulation
TP53, PMF1-BGLAP, RN7SL366P, FAM117A, SPINK8	Y chromosome mosaic loss (Y measurement) 5 of 17 loci		GWAS GWAS	5 loci in which the genotype of Sérgio F deviates from the average of the reference population in the favourable sense (favourable = more Y dosage (less Y loss)). Mechanisms: Somatic / Tissue mosaicism; Clonal hematopoiesis (CHIP)

5. Unfavourable

Gene/score	SNP	Genotype	Source · evidence	Effect
TERT	rs2736100 Intron 2 (TERT)	CC 2× C	chip Strong	Allele C: more risk of myeloproliferative neoplasms and clonal expansions. Mechanisms: Exhaustion of Stem Cells; Clonal hematopoiesis (CHIP); Hayflick's Telomeres and Limit
polygenic	G_TL Telomere length (87 loci)	the 25th percentile	86 variants Score	Low percentile = less replicative reserve (but also less risk of myeloid clones). Mechanisms: Exhaustion of Stem Cells; Hayflick's Telomeres and Limit
CD164, ITPR2-AS2	Clonal hematopoiesis (CHIP) 2 of 23 loci		GWAS GWAS	2 loci in which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = less risk of CHIP). Mechanisms: Clonal hematopoiesis (CHIP); Exhaustion of Stem Cells; Somatic Mutation Accumulation
STARD4-AS1, NPAT	Y chromosome mosaic loss (Y measurement) 2 of 17 loci		GWAS GWAS	2 loci in which the genotype of Sérgio F deviates from the mean of the reference population in the unfavourable sense (favourable = more Y dosage (less Y loss)). Mechanisms: Somatic / Tissue mosaicism; Clonal hematopoiesis (CHIP)

6. Rare variants (ClinVar) and chip boundaries

All 10 045 variants classified as pathogenic or probably pathogenic in ClinVar were searched for in the 32 genes of the panel of this report (ANKRD26, ASXL1, ATM, CEBPA, CHEK2, CXCR4, DDX41, DKC1, DNMT3A, ELane, ERCC6L2, ETV6, FANCA, FANCC, GATA2, HAX1, JAK2, MECOM, MPL11, RPL5, RPS19, RTEL1, RUNX1, SAMD9, SAMD9L, SBDS, SRP72, TERC, TERT, TET2, TP53). Of these, 646 have probes on the chip and additional 8 are in the imputation panel.

Gene	Variant	Genotype	Condition (ClinVar)	Evaluation
SBDS	rs120074160 7:66994286 T>A	AA	Shwachman syndrome; Aplastic anemia; Shwachman-Diamond syndrome 1; SBDS-related disorder; Inborn genetic diseases; not provided	Homozygosity would be Shwachman-Diamond syndrome (middle marrow failure since childhood). SBDS has an almost identical pseudogene (SBDSP1). Artifact; the imputation only repeats the chip and is not independent confirmation.
ATM	rs780905851 11:108345889 T>G	GG	Heredity cancer-predisposing syndrome; Ataxia-telangiectasia syndrome; Familial cancer of breast; not provided; ATM-report	Homozygosity for a pathogenic variant of ATM would cause ataxia-telangiectasia from childhood: incompatible with a healthy adult. Artifact of the probe (already indicated in the analysis of the raw DNA).

Chip limit: the chips genotype well common variants, but often fail in the rare ones (false positives and false negatives; Weedon 2021). The absence of pathogenic variants here does not exclude hereditary syndromes. Only clinical sequencing (panel, exome or genome) in a certified laboratory does so.

7. GWAS loci

For each study: independent loci (one SNP per region of 500 kb, the most significant, $p \leq 5 \times 10^{-8}$), with the genotype of Sérgio F and the classification against the reference population. The typical loci are in the complete table (CSV); here we show those that move away from the average.

Clonal hematopoiesis (CHIP)

Wen S et al. 2025 (about 540 000 people). Effect allele: more risk of CHIP = unfavourable. 23 loci: 7 favourable, 2 unfavourable, 12 typical, 2 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
TRIM59-IFT80, IFT80	rs4616688	T ↑	GG	0.51	imputation (DR2=0.99)	Favourable
CD164	rs11753645	C ↓	TT	0.31	imputation (DR2=0.99)	Unfavourable
POGLUT3	rs11212676	A ↑	GG	0.44	imputation (DR2=0.96)	Favourable
ITPR2-AS2, ITPR2	rs1021797	C ↑	CA	0.06	imputation (DR2=0.99)	Unfavourable
UBE2G1 - SPNS3	rs73332852	T ↑	CC	0.02	imputation (DR2=0.88)	Favourable
RUNX1	rs9975188	C ↑	TT	0.33	imputation (DR2=0.99)	Favourable
ZNF318 - ABCC10	rs7742789	T ↑	CC	0.34	imputation (DR2=1.00)	Favourable
ACTG1P25, ACTG1P25	rs12404456	T ↓	TT	0.48	imputation (DR2=0.97)	Favourable
PARP11-AS1 - LINC03141	rs582975	T ↓	TT	0.39	imputation (DR2=0.90)	Favourable

Y chromosome mosaic loss (Y measurement)

Wright DJ et al. 2017 (67 034 men). Effect allele: more Y dosage (less Y loss) = favourable. 17 loci: 5 favourable, 2 unfavourable, 7 typical, 3 without data.

Gene (mapped)	SNP	Allele	Genotype	Freq.	Source	Class
TP53	rs78378222	G ↓	TT	0.01	chip	Favourable
PMF1-BGLAP, PMF1	rs2736609	T ↓	CC	0.35	imputation (DR2=0.99)	Favourable
RN7SL366P - C6orf118	rs381500	C ↓	AA	0.52	imputation (DR2=0.99)	Favourable
STARD4-AS1, NREP	rs56084922	G ↓	GA	0.07	chip	Unfavourable
NPAT	rs4754301	A ↓	AA	0.61	imputation (DR2=1.00)	Unfavourable
FAM117A	rs77522818	A ↓	AT	0.94	imputation (DR2=0.98)	Favourable
SPINK8 - FBXW12	rs115854006	C ↓	CT	0.96	imputation (DR2=0.96)	Favourable

8. Map for 84 aging mechanisms

Mechanism	Favourable	Unfavourable	Interpretation
Clonal hematopoiesis (CHIP)	JAK2 rs12343867; TERT rs2736100; SH2B3 rs3184504; CHEK2 rs17879961; TP53 rs78378222; G_MLOG; Clonal hematopoiesis (CHIP); Y chromosome mosaic loss (Y measurement)	TERT rs2736100; Clonal hematopoiesis (CHIP); Y chromosome mosaic loss (Y measurement)	Favourable to typical (JAK2 46/1 absent, SH2B3, CHIP score at the 30th percentile).
Exhaustion of Stem Cells	JAK2 rs12343867; TERT rs2736100; SH2B3 rs3184504; Clonal hematopoiesis (CHIP)	TERT rs2736100; G_TL; Clonal hematopoiesis (CHIP)	Mixed: favourable clonality; shorter telomeres.
Hayflick's Telomeres and Limit	TERT rs2736100	TERT rs2736100; G_TL	the 25th percentile (light disadvantage).
Inflammaging	SH2B3 rs3184504	—	SH2B3 CC (favourable).
DNA Damage and DNA Repair Deficit	CHEK2 rs17879961	—	Predominant
Somatic / Tissue Mosaicism	TP53 rs78378222; G_MLOG; Y chromosome mosaic loss (Y measurement)	Y chromosome mosaic loss (Y measurement)	Low predicted Y loss (favourable).
Somatic Mutation Accumulation	Clonal hematopoiesis (CHIP)	Clonal hematopoiesis (CHIP)	No plausible rare variants.

9. Practical section — suggestion (not applied)

Coherent with the existing genetic plan and the current stack reported. Nothing was applied here; everything is to discuss with the doctor.

#	Suggestion	Why (genes)	State
1	Reduce chronic inflammation (more fiber, 3 to 3.2-4 g/day of EPA+DHA, sleep)	Inflammation (IL-6, IL-1 β) gives advantage to clones DNMT3A/TET2.	suggestion (not applied)
2	LDL and ApoB as low as possible (lower is always better)	CHIP is added to cardiovascular risk through plaque inflammation.	suggestion (not applied)
3	Regular exercise	It maintains the function of muscle and hematopoietic stem cells.	suggestion (not applied)
4	Maintaining intermittent rapamycin (already in the stack)	In models, intermittent inhibition of mTOR preserves the quiescence of stem cells.	suggestion (not applied)
5	Hemogram with RDW and, at 50+ or if changes arise, sequencing for CHIP	It detects an early clonal expansion.	suggestion (not applied)
6	Avoiding radiation and unnecessary chemotherapy	They are the largest accelerators of clone selection with TP53/PPM1D.	suggestion (not applied)

Avoiding or with caution

- Activators of telomerase (TA-65, astral; pause the epitalon is not applied) with TERT rs2736100 CC. (*suggestion, not applied*)
- Unvalidated stem cell therapies and growth factors or GH without medical indication. (*suggestion, not applied*)
- Tobacco smoke (the largest environmental factor of CHIP). (*suggestion, not applied*)

10. Limitations

- CHIP is acquired: only the sequencing of blood to detect; the scores measure predisposition.
- The regenerative capacity of solid tissues does not have good common genetic markers.
- The GWAS loci show average associations in the population; the "mapped" gene is the closest and not always the causal gene.
- The effects of isolated SNPs are small; several studies with only one population. They do not allow clinical conclusions by themselves.
- Imputation: an imputed genotype may be wrong, especially in rare variants; the imputed ones are marked.
- This report uses only genes: analysis, exams or measurements were not used.

11. Full table SNP to SNP (cured SNPs)

Gene	SNP	Effect allele (+)	Genotype	Freq.	Source	Chain	Class	Evidence
TCL1A	rs2887399 TCL1A promoter	T (protector)	GG	0.206	chip	strand + (without conversion)	Typical	Moderate
JAK2	rs12343867 haplotype 46/1 (GGCC)	C (risk)	TT	0.276	chip	strand + (without conversion)	Favourable	Strong
TERT	rs2736100 Intron 2 (TERT)	C (commitment)	CC	0.506	chip	strand + (without conversion)	Commitment	Strong
TERT	rs7705526 Intron 2 (TERT)	A (commitment)	AC	0.346	chip	strand + (without conversion)	Typical	Strong
SH2B3	rs3184504 Trp262Arg (LNK)	T (risk)	CC	0.461	chip	strand + (without conversion)	Favourable	Moderate
CHEK2	rs17879961 I157T	G (risk)	AA	0.004	chip	strand + (without conversion)	Favourable	Moderate
TP53	rs78378222 polyadenylation signal	G (risk)	TT	0.013	chip	strand + (without conversion)	Favourable	Moderate

Notes about the typical, informative and without data: TCL1A rs2887399 (GG): The T allele reduces the activation of TCL1A in stem cells with mutations of DNMT3A/TET2 and decreases the clonal advantage: minus CHIP (Weinstock 2023). TERT rs7705526 (AC): Allele A: longer telomeres.